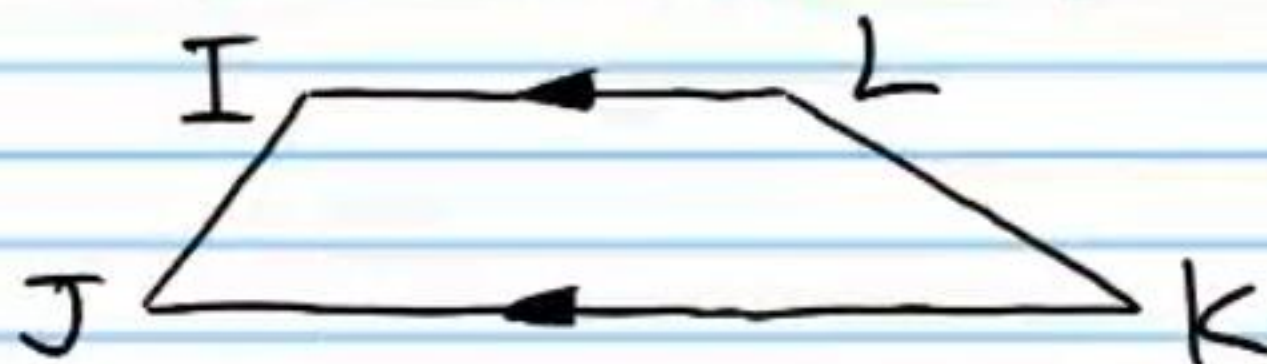


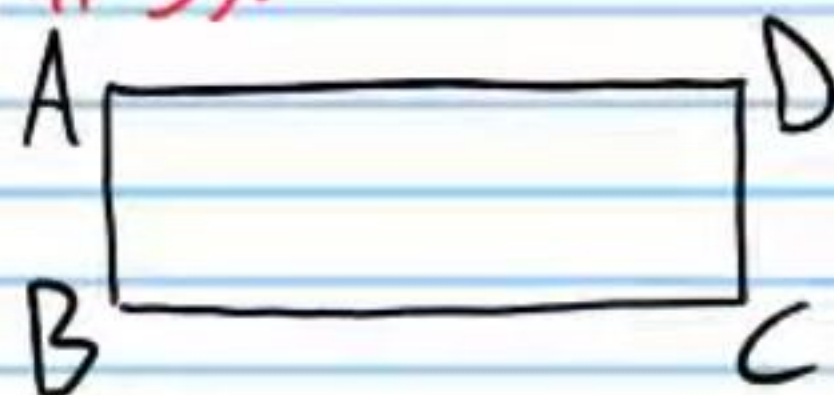
Special Quadrilaterals (Part III): More Proofs (Homework)

Write a two-column proof for each statement below. You will need to reference the definitions of the special quadrilaterals.

- ① If the quadrilateral IJKL below is a trapezoid, then $\angle J$ and $\angle I$ are supplementary (this is the Trapezoid Non-Base Pair Theorem). Hint: You will need the Consecutive Interior Angles Theorem.



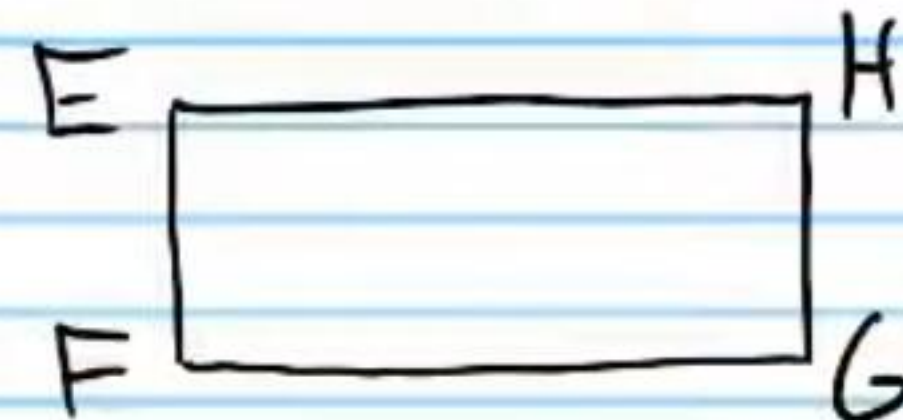
- ② If the quadrilateral ABCD below is a rectangle, then opposite sides are congruent (this is the Rectangle Opposite Sides Theorem). Hint: You must use the Rectangle Parallelogram Theorem (proved in class problem #5).



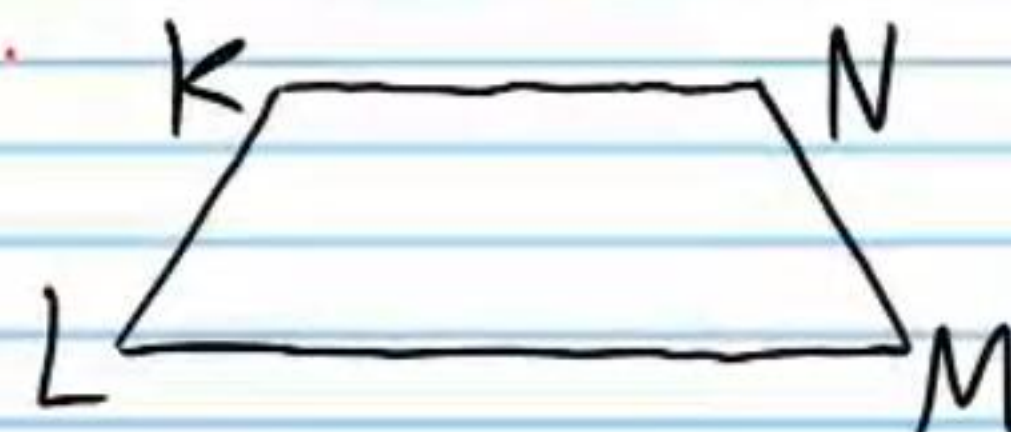
- ③ If the quadrilateral EFGH below is a kite, then $\angle E \cong \angle G$ (this is the Kite Congruent Angles Theorem). Hint: You must draw a diagonal.



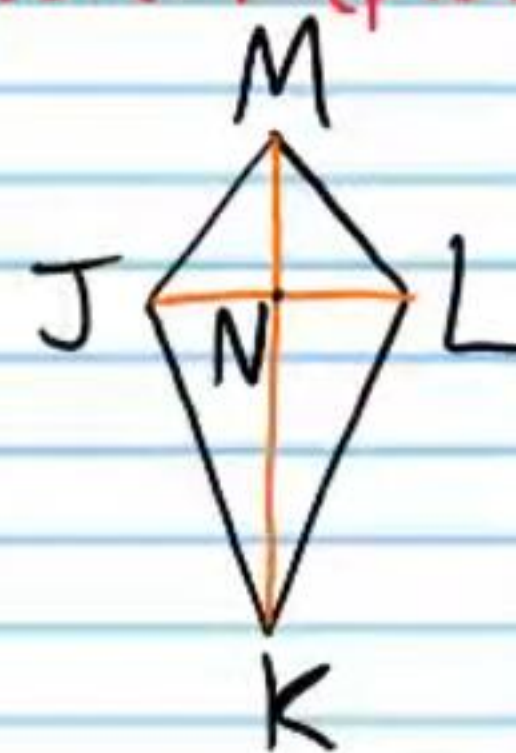
- ④ If the quadrilateral EFGH below is a rectangle, then its diagonals bisect each other (this is the Rectangle Diagonal Bisection Theorem). Hint: You must use the Rectangle Parallelogram Theorem (proved in class problem #5).



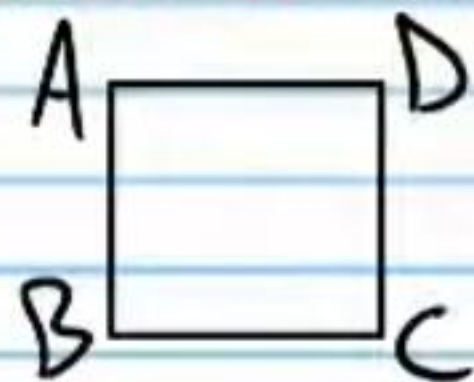
- ⑤ If the trapezoid KLMN below is isosceles, then its diagonals are congruent (this is the Trapezoid Diagonals Theorem). Hint: Use the Trapezoid Congruent Base Angles Theorem (proved in class problem #11) and draw the diagonals one at a time.



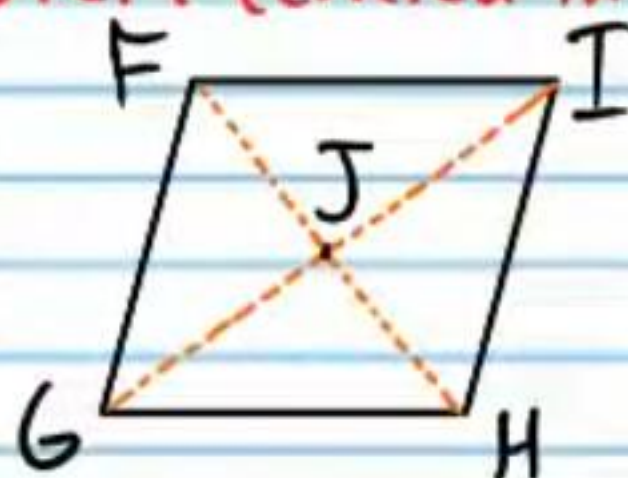
- ⑥ If the quadrilateral JKLM below is a kite, then MK bisects JL (this is the Kite Diagonal Bisector Theorem). Hint: You will need to use the Kite Angle Bisector Theorem (proved in class problem #4).



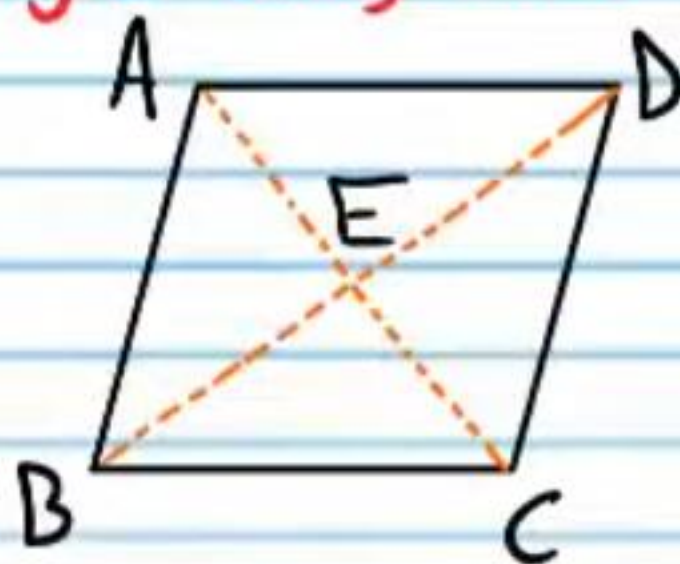
⑦ If the quadrilateral ABCD below is a square, then it is a parallelogram (this is the Square Parallelogram Theorem). Hint: Reference in-class problem #5.



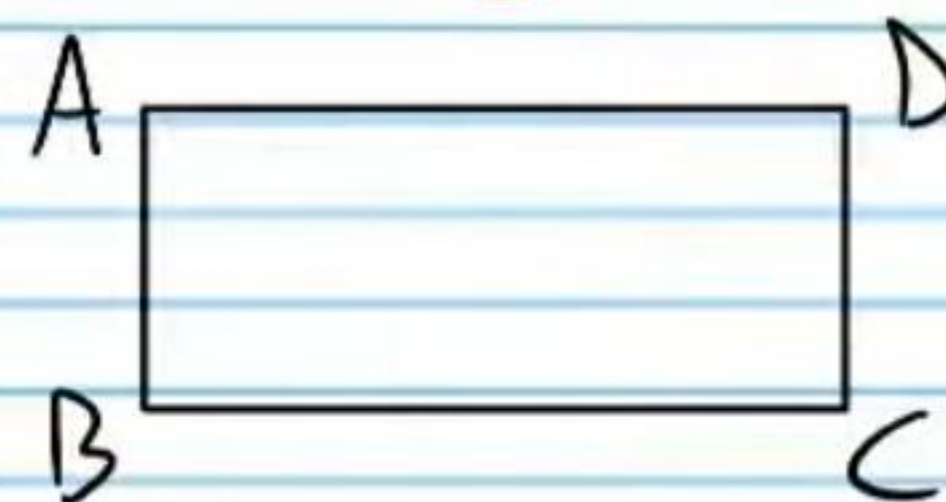
⑧ If the diagonals of the parallelogram FGHI below are perpendicular, then it is a rhombus (this is the converse of a slightly modified version of the Rhombus Perpendicular Diagonals Theorem). Hint: Use a parallelogram theorem (it is given that the figure is a parallelogram), and the Right Angle Congruence Theorem (covered in a previous class).



⑨ If the quadrilateral ABCD below is a rhombus, then its diagonals divide it into four congruent right triangles (this is the Rhombus Right Triangles Theorem). Hint: Use the Rhombus Perpendicular Diagonals Theorem (proved in class problem #8), the Rhombus Parallelogram Theorem (proved in class problem #6), and the Right Angle Congruence Theorem (a previous class).



⑩ If the diagonals of the parallelogram ABCD below are congruent, then it is a rectangle (this is the converse of a slightly modified version of the Rectangle Diagonals Theorem). Hint: Use $\triangle ABC$ and $\triangle ABD$ and the Consecutive Interior Angles Theorem. It is given that ABCD is a parallelogram.

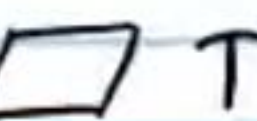
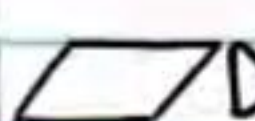


Homework Answers

①	Statement	Reason
I)	$IJKL$ is a trapezoid.	Given.
II)	$\overline{IL} \parallel \overline{JK}$.	Def of  .
III)	$\angle I$ & $\angle J$ are supplementary.	CIAT

②	Statement	Reason
I)	$ABCD$ is a rectangle.	Given.
II)	$ABCD$ is a parallelogram.	 -  Thm.
III)	$\overline{AD} \cong \overline{BC}$ & $\overline{AB} \cong \overline{DC}$.	 Opp. Sides Thm.

③	Statement	Reason
I)	$EFGH$ is a kite.	Given.
II)	$\overline{HF} \cong \overline{HF}$	Ref. Prop of $\cong$.
III)	$\overline{EH} \cong \overline{HG}$ and $\overline{EF} \cong \overline{FG}$	Def. of kite.
IV)	$\triangle FEH \cong \triangle FGH$	SSS
V)	$\angle E \cong \angle G$	CPCTC

④	Statement	Reason
I)	$EFGH$ is a rectangle.	Given.
II)	$EFGH$ is a parallelogram.	 -  Thm.
III)	$\overline{EG}$ & $\overline{FH}$ bisect each other.	 Diagonals Thm.

⑤	Statement	Reason
I)	$KLMN$ is an isosceles trapezoid.	Given.
II)	$\overline{KL} \cong \overline{NM}$.	Def of isosceles  .
III)	$\angle L \cong \angle M$	 $\cong$ Base $\angle$'s Thm.
IV)	$\overline{LM} \cong \overline{LM}$	Ref. Prop of $\cong$.
V)	$\triangle LKM \cong \triangle MNL$.	SAS
VI)	$\overline{KM} \cong \overline{NL}$	CPCTC

⑥	Statement	Reason
I)	$JKLM$ is a kite.	Given.
II)	$\overline{JM} \cong \overline{ML}$.	Def. of a kite.

III) $\overline{MN} = \overline{MN}$	Ref. Prop of $\cong$.
IV) $\angle JMN \cong \angle LMN$	Kite $\angle$ Bisector Thm.
V) $\triangle JMN \cong \triangle LMN$	SAS
VI) $\overline{JN} \cong \overline{NL}$	CPCTC
VII) $\overline{MK}$ bisects $\overline{JL}$	Def of Seg. bisector.

Statement	Reason
I) ABCD is a square.	Given.
II) $\angle A$, $\angle B$, and $\angle D$ are rt. $\angle$'s.	Def. of $\square$.
III) $m\angle A = 90^\circ$, $m\angle B = 90^\circ$, and $m\angle D = 90^\circ$	Def. of rt. $\angle$.
IV) $m\angle A + m\angle B = 90 + 90$ and $m\angle A + m\angle D = 90 + 90$.	Subs. Prop of $=$.
V) $m\angle A + m\angle B = 180$ $m\angle A + m\angle D = 180$	Simplification.
VI) $\angle A$ & $\angle B$ are supplementary $\angle$'s and $\angle A$ & $\angle D$ are supplementary $\angle$'s.	Def. of Supp. $\angle$'s.
VII) $\overline{AD} \parallel \overline{BC}$ & $\overline{AB} \parallel \overline{CD}$.	CIATC
VIII) ABCD is a $\square$.	Def. of $\square$.

Statement	Reason
I) FGHI is a parallelogram.	Given.
II) $\overline{FH}$ & $\overline{GI}$ bisect each other.	$\square$ Diagonals Thm.
III) $\overline{FJ} \cong \overline{JH}$ & $\overline{GJ} \cong \overline{JI}$	Def. of seg bisector.
IV) $\overline{FH} \perp \overline{GI}$	Given.
V) $\angle FJG$, $\angle GJH$, $\angle FJI$, and $\angle IJH$ are right angles.	Def of $\perp$.
VI) $\angle FJG \cong \angle GJH \cong \angle FJI \cong \angle IJH$	Rt $\angle \cong$ Thm.
VII) $\triangle FJG \cong \triangle HJG \cong \triangle HJI \cong \triangle FJI$	SAS
VIII) $\overline{FG} \cong \overline{FI} \cong \overline{GH} \cong \overline{HI}$.	CPCTC
IX) FGHI is a rhombus.	Def of Rhombus.

Statement	Reason
I) ABCD is a rhombus.	Given.
II) $\overline{AC} \perp \overline{BD}$.	Rhombus $\perp$ Diagonals Thm.
III) $\angle AEB$, $\angle AED$, $\angle DEC$, and $\angle BEC$ are rt. $\angle$'s.	Def. of $\perp$.
IV) $\triangle AEB$, $\triangle AED$, $\triangle CED$, and $\triangle CEB$ are right triangles.	Def of Rt. $\triangle$

V) $\angle AEB \cong \angle AED \cong \angle DEC \cong \angle BEC$	Rt. $\angle \cong$ Thm.
VI) ABCD is a parallelogram.	Rhombus $\square$ Thm.
VII) $\overline{AC}$ & $\overline{BD}$ bisect each other.	$\square$ Diagonals Thm.
VIII) $\overline{AE} \cong \overline{EC}$ & $\overline{BE} \cong \overline{ED}$.	Def. of seg. bisector.
IX) $\triangle AEB \cong \triangle AED \cong \triangle CED \cong \triangle CEB$	SAS

⑩

Statement	Reason
I) ABCD is a parallelogram.	Given.
II) $\overline{AD} \cong \overline{BC}$.	$\square$ Opposite Sides Thm.
III) $\overline{AC} \cong \overline{BD}$	Given.
IV) $\overline{AB} \cong \overline{AB}$	Ref. Prop of $\cong$.
V) $\triangle ABD \cong \triangle BAC$.	SSS
VI) $\angle A \cong \angle B$	CPCTC
VII) $m\angle A = m\angle B$	Def of $\cong$ $\angle$'s.
VIII) $\angle A$ & $\angle B$ are supplementary.	CIAT
IX) $m\angle A + m\angle B = 180^\circ$	Def. of Supp. $\angle$'s.
X) $m\angle A + m\angle A = 180^\circ$	Subs. Prop of $=$.

XI) $2m\angle A = 180^\circ$	Simplification.
XII) $m\angle A = 90^\circ$	Div. Prop of $=$.
XIII) $m\angle B = 90^\circ$	Trans. Prop of $=$.
XIV) $\angle A \cong \angle C$ & $\angle B \cong \angle D$	$\square$ Opposite $\angle$'s Thm.
XV) $m\angle A = m\angle C$ & $m\angle B = m\angle D$	Def of $\cong$ $\angle$'s
XVI) $m\angle C = 90$ & $m\angle D = 90$.	Trans. Prop of $=$.
XVII) $\angle A, \angle B, \angle C,$ and $\angle D$ are right angles.	Def of rt. $\angle$.
XVIII) ABCD is a rectangle.	Def of $\square$.